

MPC-PiNNs technique applied to Quadrotors

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Todo list

outline of the work	2
unicycle approach	2
uav apporach	2
uav simplified model	2
uav complete model	2
mpc structure	2
losses	3
pipeline	3
image pinn	3
what we have obtained	3
problems	3
pinn vs pinnc	3

Contents

Todo list	1
1 Introduction	2
2 Model Predictive Control	2
3 Physical Informed Neural Network	2
4 Results	3
5 Conclusions	4

1 Introduction

outline of the work

unicycle approach

uav approach

Model Predictive Control (MPC) has emerged as a powerful control strategy in robotics due to its ability to handle constraints and optimize system behavior over a prediction horizon. When combined with neural networks, which can learn complex system dynamics, it creates a promising framework for controlling nonlinear systems.

This report presents the implementation and analysis of a Neural Model Predictive Control (NMPC) scheme applied to a quadrotor. The unicycle serves as an excellent test case due to its nonlinear dynamics, making it both challenging and representative of real-world robotic systems. Besides, we followed a bottom-up approach, testing the pipeline first on a unicycle, which still has a nonlinear dynamics, but is easier to treat due a simpler model and faster prototyping possibilities.

Our approach integrates deep learning techniques to approximate the quadrotor's dynamics, which are then utilized within an MPC framework to generate optimal control inputs. We explore the effectiveness of this method in trajectory tracking tasks, comparing it with traditional control approaches.

The following sections detail the theoretical foundations, implementation specifics, and experimental results of our neural MPC system. We also discuss the challenges encountered and potential improvements for future work.

2 Model Predictive Control

uav simplified model

uav complete model

mpc structure

3 Physical Informed Neural Network

Physical Informed Neural Networks (PiNNs) are a recent approach that exploit physical laws inside the learning process of a Neural Network. As previously stated, we have used a bottom-up approach by first testing this approach tu a Unicycle. This robot still has a nonlinear dynamics but it's way simpler than the quadrotor one. Its configuration vector is composed by the cartesian position (x, y) and its

orientation angle θ .

$$\begin{cases} \dot{x} = v \cos(\theta) \\ \dot{y} = v \sin(\theta) \\ \dot{\theta} = w \end{cases} \quad (1)$$

$$\mathbf{R}_{\text{RPY}} = \begin{bmatrix} \cos \psi \cos \theta & \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi \\ \sin \psi \cos \theta & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix} \quad (2)$$

$$\dot{\mathbf{v}} = \begin{bmatrix} 0.0 \\ 0.0 \\ -g \end{bmatrix} + \mathbf{R}_{\text{RPY}} \cdot \begin{bmatrix} 0.0 \\ 0.0 \\ \frac{T}{m} \end{bmatrix} \quad (3)$$

$$\begin{cases} \dot{x} = v_x \\ \dot{y} = v_y \\ \dot{z} = v_z \\ \dot{v}_x = (\cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi) \frac{T}{m} \\ \dot{v}_y = (\sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi) \frac{T}{m} \\ \dot{v}_z = (\cos \theta \cos \phi) \frac{T}{m} - g \\ \dot{\phi} = \\ \dot{\theta} = \\ \dot{\psi} = \end{cases} \quad (4)$$

In our implementation, we have used a PiNNs to predict the next state of the system, given the control input obtained from the Model Predictive Control.

losses

pipeline

image pinn

4 Results

what we have obtained

problems

pinn vs pinnc

5 Conclusions